

MANISTIQUE, MI, USA

WMO: 726399

Lat: 45.950N

Lon: 86.230W

Elev: 178

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 86-04

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.6	-16.0	-24.4	0.4	-16.3	-22.2	0.5	-15.3	14.5	-4.4	12.6	-4.3	4.0	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	25.2	18.3	23.8	18.3	22.6	17.8	21.1	23.5	20.2	22.5	19.2	21.4	3.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.1	22.9	19.0	14.1	21.8	18.0	13.2	20.6	61.9	23.6	58.5	22.6	55.5	21.3	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.9	9.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-5.5	-6.2	-2.0	3.0	9.0	14.4	17.8	18.3	14.9	7.7	2.4	-1.9
	DBStd	9.40	5.15	4.79	4.58	3.10	3.09	3.10	2.56	2.80	3.86	3.59	3.70	5.15
	HDD10.0	2273	480	453	371	211	56	3	0	0	7	90	230	371
	HDD18.3	4556	738	687	629	461	291	122	41	36	113	330	479	629
	CDD10.0	831	0	0	0	0	24	136	242	256	154	18	1	0
	CDD18.3	73	0	0	0	0	0	5	25	33	9	0	0	0
	CDH23.3	171	0	0	0	0	3	16	58	89	4	0	0	0
	CDH26.7	12	0	0	0	0	0	4	3	6	0	0	0	0
Wind	WSAvg	4.2	5.0	4.5	4.5	4.3	3.8	3.2	3.1	3.7	4.1	4.7	5.0	4.7
Precipitation	PrecAvg	780	44	31	50	64	78	78	77	75	88	81	63	48
	PrecMax	1003	144	81	124	123	142	172	171	113	207	148	122	100
	PrecMin	620	17	4	8	20	41	25	16	22	29	31	24	6
	PrecStd	96	27	18	32	26	26	35	41	21	36	33	27	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	4.2	8.5	13.6	21.6	25.1	26.9	27.4	23.7	17.4	11.5	11.3
		MCWB	1.6	1.2	4.9	7.4	12.6	17.6	18.0	20.5	19.2	14.7	9.6	2.9
	2%	DB	3.8	2.6	6.0	10.6	17.5	22.8	25.0	25.7	22.4	15.6	9.9	7.3
		MCWB	0.3	0.7	2.8	6.2	11.6	17.0	18.0	19.4	18.9	12.7	8.2	1.9
	5%	DB	2.2	1.5	4.2	8.6	15.3	21.0	23.6	24.0	21.3	14.1	8.5	5.0
		MCWB	0.4	-0.1	1.7	5.3	10.6	16.0	18.1	18.6	18.1	11.6	6.9	2.3
	10%	DB	1.2	0.5	2.9	7.4	13.6	19.2	22.4	22.8	20.2	13.0	7.6	3.8
		MCWB	-0.2	-1.0	0.9	4.6	9.4	15.1	17.4	17.9	17.4	10.5	6.1	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	1.8	5.2	9.0	15.0	19.5	21.9	22.7	21.2	15.2	10.1	4.6
		MCDB	5.7	3.3	7.9	11.8	19.3	23.4	24.3	26.0	22.5	17.1	11.3	6.5
	2%	WB	1.3	0.8	3.3	7.3	12.7	18.0	20.9	21.4	19.8	13.5	8.6	3.5
		MCDB	3.0	2.3	5.1	9.4	16.0	21.0	23.2	24.2	21.3	15.1	9.4	5.5
	5%	WB	0.6	0.1	2.2	6.0	11.3	16.8	19.8	20.4	18.9	12.1	7.6	2.7
		MCDB	2.1	1.5	3.6	8.1	14.4	19.8	22.1	22.9	20.4	13.7	8.2	4.8
	10%	WB	0.0	-0.8	1.4	4.8	10.1	15.6	18.8	19.5	17.9	10.8	6.4	1.7
		MCDB	1.5	0.4	2.6	7.2	13.1	18.9	21.2	21.9	19.7	12.8	7.7	3.8
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.5	6.9	6.5	7.7	7.7	7.7	7.7	7.0	6.0	5.4	5.6
		MCDBR	7.5	7.5	7.2	8.1	10.1	9.6	9.1	8.5	6.8	6.6	5.6	5.6
		MCWBR	6.1	6.1	5.5	5.5	6.3	5.9	5.6	5.5	4.6	5.2	5.2	4.2
	5% WB	MCDBR	6.6	7.3	6.3	7.4	9.2	8.2	7.3	7.4	6.0	6.1	5.6	5.3
		MCWBR	5.7	6.0	5.1	5.4	6.2	5.7	5.3	5.3	4.7	5.3	5.3	4.6
Clear-Sky Solar Irradiance	taub	0.269	0.282	0.311	0.355	0.384	0.398	0.394	0.388	0.369	0.335	0.303	0.272	
	taud	2.403	2.343	2.378	2.362	2.347	2.342	2.368	2.385	2.436	2.511	2.515	2.449	
	Ebn at Noon	846	901	920	903	883	869	868	860	847	832	800	804	
	Edh at Noon	70	91	102	115	121	122	117	111	97	78	64	61	
All-Sky Solar Radiation	RadAvg	1.19	2.11	3.45	4.54	5.57	6.01	6.13	5.19	3.91	2.32	1.27	0.91	
	RadStd	0.11	0.25	0.33	0.46	0.46	0.43	0.46	0.28	0.34	0.29	0.26	0.15	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page